TEXORA · LECTURE SHEET

Process Control in Mixing

The mixing is the mill's recipe: the proportions of different cottons (or fibres) laid down together decide yarn strength, evenness, dyeing behaviour and cost before a single machine runs. This sheet covers bale selection by measured fibre properties, the lay-down, weighted-average calculations, and the controls that keep every lay-down identical to the recipe.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 2 · SPECIALIZATION — YARN MANUFACTURING

1. Why mixing is a control point, not a formality

No single cotton lot is uniform, and no lot is large enough to spin for long. A mixing blends several lots so that their individual deviations cancel and the blend's average properties sit on specification. If the mixing changes — because a lot ran out and was replaced casually — the yarn changes with it: strength drops, count CV rises, and dyed fabric shows barré from micronaire-driven dye-uptake differences between lots. Mixing control therefore means: the *recipe* is fixed by measured fibre properties, and the *lay-down* is executed bale by bale exactly as the recipe states.

2. Selecting bales by measured properties

Modern bale management tests every bale (HVI: length, uniformity, strength, micronaire, colour, trash) and groups bales into categories. A mixing is then drawn from categories in fixed proportions so that the blend averages stay constant even as individual bales change.

PROPERTY	WHY IT IS CONTROLLED	TYPICAL BLEND TARGET LOGIC
Staple length & uniformity	Sets spinnable count and drafting behaviour	Blend mean held within ±0.5 mm of recipe
Fibre strength	Main lever on yarn strength	Weak lots balanced by strong lots
Micronaire (fineness/maturity)	Affects dye uptake, neps, ends down	Narrow range in blend; avoid mixing extremes
Trash %	Loads the blowroom and card	Weighted trash known before the line runs

Colour grade	Visible in grey fabric and after dyeing	Categories kept consistent within a mix run

3. The weighted-average calculations

Every blend property and the blend cost are mass-weighted averages of the components:

$$\text{Blend value} = \Sigma\ (\text{proportion}_i \times \text{value}_i)$$

WORKED EXAMPLE — BLEND COST AND TRASH

Recipe: cotton A 55% (cost 2.60 units/kg, trash 3.2%), cotton B 30% (2.20 units/kg, trash 4.1%), cotton C 15% (1.90 units/kg, trash 5.0%).

Blend cost = 0.55 × 2.60 + 0.30 × 2.20 + 0.15 × 1.90 = 1.430 + 0.660 + 0.285 = **2.375 units/kg**.

Blend trash = 0.55 × 3.2 + 0.30 × 4.1 + 0.15 × 5.0 = 1.76 + 1.23 + 0.75 = **3.74%**.

A lay-down of 40 bales × 170 kg = 6800 kg therefore costs 6800 × 2.375 = **16,150 units**, and the blowroom can expect about 3.74% trash in its feed — the figure its cleaning efficiency (YM-3207, Topic 1) will be judged against.

4. The lay-down

The proportions are executed physically: bales are arranged in a lay-down in which each component occupies its recipe share of the plucker's path, so every pass of the bale plucker takes the components in the right ratio.

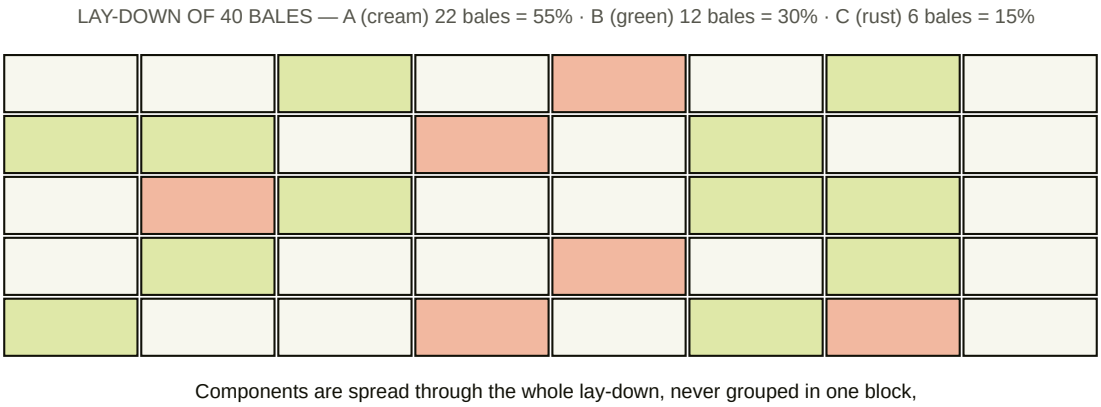


Figure 1 — A 40-bale lay-down in the recipe proportions 55/30/15. Because the plucker traverses the whole lay-down, distributing each component evenly through it makes every traverse a miniature of the recipe.

Control checks at the lay-down: bale identities and counts verified against the recipe sheet before the line starts; no substitution without a recalculated recipe; and the lay-down size matched to consumption

so a mixing runs long enough to be worth its stability (a mixing that lasts only a few hours creates a transition in the yarn every few hours).

5. Mixing faults and their signatures

- **Sudden strength fall across all frames** — a weak component proportion has crept up, or a lot was replaced by a shorter/weaker one.
- **Dye barré with good grey fabric** — micronaire spread inside the blend too wide; the yarns differ only after dyeing.
- **Gradual drift over days** — the lay-down is being "topped up" with whatever bale is nearest, so proportions wander from the recipe.
- **High nep level from the blowroom onward** — immature fibre (low micronaire component) too high in the blend.

For man-made fibre blends the same control applies with blend ratio verified by chemical analysis of the yarn: a polyester/cotton blend sold as 65/35 must test within its tolerance, and the ratio is set here, at the mixing, not corrected anywhere later.

Glossary

Mixing — the fixed recipe of fibre components and their proportions for a yarn. **Lay-down** — the physical arrangement of bales from which the plucker feeds. **Bale management** — selecting bales by tested properties so blend averages stay constant. **Micronaire** — an airflow measure combining fineness and maturity of cotton. **HVI** — High Volume Instrument, the rapid bale-testing system. **Barré** — visible stripes in dyed fabric from yarn property differences. **Component** — one cotton lot or fibre type within a mixing.

Key points

• The mixing fixes yarn properties and cost before spinning starts; it is controlled by measured bale data, not by eye.
• Blend properties and cost are mass-weighted averages: cost 2.375 units/kg and trash 3.74% in the example.
• The lay-down executes the recipe physically — components distributed evenly, counts verified, no casual substitution.
• Mixing faults have recognisable signatures: strength falls, dye barré, day-to-day drift, high neps.
• Blend ratios in synthetic blends are verified by analysis of the yarn, and are set only here.

Practice

1. A recipe uses cotton X 60% (strength 30 g/tex) and cotton Y 40% (strength 26 g/tex). Calculate the expected blend strength index, and the blend cost if X costs 2.80 and Y 2.30 units/kg.
2. A lay-down is planned as 30 bales. The recipe is 50/30/20. How many bales of each component are needed, and what is the lay-down mass at 170 kg per bale?
3. A mill replaces an exhausted component with a cheaper cotton of the same staple length but much lower micronaire. Predict two problems that may appear, and where each will first be detected.

Answer key

1. Strength index = 0.60 × 30 + 0.40 × 26 = 18.0 + 10.4 = **28.4 g/tex**. Cost = 0.60 × 2.80 + 0.40 × 2.30 = 1.68 + 0.92 = **2.60 units/kg**.
2. Bales: 0.50 × 30 = **15**, 0.30 × 30 = **9**, 0.20 × 30 = **6**. Mass = 30 × 170 = **5100 kg**.
3. (1) Nep level and ends down rise — immature fibres break and roll into neps; first detected at card sliver nep tests and ring-frame ends-down records. (2) Dye-uptake differences cause barré after dyeing; the grey yarn may test normally, so it is first detected at fabric dyeing unless blend micronaire is monitored at the mixing.